MIL-T-36160A(DM)
17 March 1965
SUPERSEDES
MIL-T-36160 (BuMed)
31 July 1963

MILITARY SPECIFICATION

# TUBE, ENDOTRACHEAL, MAGILL (PLASTIC)

## 1. SCOPE

1.1 Scope. This specification covers endotracheal tubes (intratracheal catheters), of the Magill design, suitable for use in anesthetic procedures.

1.2 Classification. Tubes shall be of the following sizes, as specified (see 6.1):

Size - 5.5 mm. I.D.
Size - 7.0 mm. I.D.
Size - 8.0 mm. I.D.
Size - 8.5 mm. I.D.
Size - 9.0 mm. I.D.
Size - 9.5 mm. I.D.
Size - 10 mm. I.D.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |
| PPP-B-676 | Boxes, Set-up, Paperboard. |

FSC 6515

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Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

Fed. Test Method Std. No. 601 Rubber: Sampling and Testing.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Standards Association, Incorporated

ASA Z79.1-1960 American Standard Specifications for Anesthetic Equipment: Endotracheal Tubes

(Application for copies should be addressed to the American Standards Association, Inc., 10 East 40th Street, New York 16, N. Y.)

U.S. Pharmacopoeial Convention, Incorporated

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa.)

3. REQUIREMENTS

3.1 Material. Shall be transparent, flexible and of a medical grade polyvinyl tubing. Shall contain not more than five (5) parts per million (p.p.m.) of lead when tested as specified in 4.4.4. Catheters shall be unaffected by oil base lubricants. Shall be sterile, pyrogen free, pass the animal implantation test and be resterilizable when tested as specified in section 4. Shall have a hardness (Shore A durometer) of 82 ± 3 when tested as specified in 4.4.6.

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3.2 Dimensions, style and design. Endotracheal tubes (intratracheal catheters) shall be of the Magill design and in accordance with applicable portions of sections 3 and 6 of ASA Standard Z79.1-1960 for the sizes indicated for oral tubes. The wall thickness for each size shall be as follows:

| Size ((I.D. (mm.)) | Wall thickness (mm.) + .15 mm. | For use with endotracheal cuff I.D. (mm.) |
|---|---|---|
| 5.5 | 1.13 | 6.5 |
| 7.0 | 1.38 | 6.5 |
| 8.0 | 1.55 | 8.0 |
| 8.5 | 1.63 | 8.0 |
| 9.0 | 1.71 | 9.5 |
| 9.5 | 1.79 | 9.5 |
| 10.0 | 1.88 | 9.5 |

3.3 Sterilization. When tested as specified in 4.4.5, tubes shall show no signs of brittleness, blistering, swelling, peeling, cracking or any other damage or deformation.

3.4 Sterility. Tubes shall be sterile when tested as specified in 4.4.2.

3.5 Freedom from pyrogen. Tubes shall be pyrogen free when tested as specified in 4.4.1.

3.6 Animal implantation. The tubes shall be nontoxic and free from tissue reaction. When subjected to the animal implantation test specified in 4.4.3, there shall be no evidence of toxicity, or adverse tissue reaction upon gross and microscopic examination.

3.7 Certificates of sterility, freedom from pyrogens, and animal implantation. A certificate(s) listing all lot or control numbers, shall state that representative sample(s) from each lot have been tested and have been found to be sterile, free from pyrogens, and have passed the animal implantation test. This certificate(s) shall include the specific tests performed and the test results. The record of the animal implantation test and its results shall be certified by the Pathologist who performed or supervised performance of the animal implantation test. For destination inspection only, one copy of certificate(s) shall be submitted by the contractor directly to Chief, Quality Control and Maintenance Division Directorate of Technical Operations, Defense Medical Supply Center, 29th Street and 3rd Avenue, Brooklyn, New York, 11232, and a copy shall be forwarded with shipment to the consignee.



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3.8 Manufacturer's record. The manufacturer shall place a code or lot number on the paper label, which can be verified against the manufacturer's records of all tests made on animal implantation, sterile, pyrogen-free tubes. These records shall be made available for inspection by any duly authorized Government representative. No shipment shall be made until all tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.9 Identification marking. Tubes shall be marked in accordance with applicable portions of section 5 of ASA Standard Z79.1-1960. In addition, each tube shall be legibly and permanently marked with manufacturer's name or registered trademark.

3.10 Workmanship. Workmanship shall be first class throughout. The finished tubes shall be symmetrical in shape, and the surface shall have a smooth finish, free from irregularities or imperfections of any kind which detract from the tubes appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer, of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.



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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. Unit of product shall be one tube.

4.2.2 For tests.

4.2.2.1 For pyrogen and sterility tests. Sampling procedures for pyrogen and sterility tests shall be conducted as indicated in the current edition of the Pharmacopeia of the United States, for transfusion and infusion assemblies.

4.2.2.2 For implantation test. Each manufacturer's lot of tubing shall be tested as specified in 4.4.3. If more than two tissue sites per animal show a reaction, the lot represented shall be rejected.

4.2.2.3 Sterilization tests. Sampling for each test shall be conducted in accordance with Standard MIL-STD-105, inspection level I2, except that the acceptance number shall be zero (0) for all sample sizes, and a minimum of four (4) samples shall be taken regardless of lot size. Unit of product shall be one tube.

4.2.2.4 For lead test. Samples shall be taken for each lot of plastic as indicated in paragraph 4.4.4.

4.2.2.5 For hardness test. Sufficient samples shall be selected from each lot to perform the test. Failure to comply with test requirements shall be cause for rejection of the lot represented.

4.2.3 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The tubes shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests.

4.4.1 Pyrogen. Tubes shall be tested for freedom from pyrogens, in accordance with the current edition of the Pharmacopeia of the United States, for transfusion and infusion assemblies.

4.4.2 Sterility. Tubes shall be tested for sterility, in accordance with the current edition of the Pharmacopeia of the United States, for transfusion and infusion assemblies.



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4.4.3 Animal implantation test. The samples selected shall have been processed (including sterilization) so as to be representative of the finished product. Care shall be exercised in the preparation of samples and specimens to prevent contamination with micro-organisms and other foreign matter. Clean cutting devices by appropriate method (e.g., successive cleaning with acetone and methylene chloride) prior to sub-dividing a sample. Clean all other equipment by thorough detergent scrubbing and prolonged rinsing with tap and distilled water. Render containers and devices used in transfer and administration of test material sterile and dry by a suitable process. If ethylene oxide, or similar agent, is used as sterilizing agent, sufficient time for complete degassing shall be allowed. Each sample (cut from manufacturer's lot of tubing) shall be divided into four specimens. The specimens shall be comprised of solid strips measuring at least 1 millimeter (mm.) by 10 mm. The edges of the strips shall be smooth to avoid mechanical trauma upon implantation. Any foreign material, such as lint and particles, shall be removed by placing the specimens in a beaker or cylinder and adding approximately 70 milliliters (ml.) of water for injection, and then agitating for approximately 30 seconds. This cleaning procedure shall be performed twice. (Note: The specimens shall not be cleaned with a cloth or by washing with an organic solvent, a surface active agent, etc.) The test shall be performed in a clean area using aseptic techniques. Healthy, adult rabbits weighing 4 to 6 pounds whose paravertebral muscles are sufficiently large in size to allow for embedding of the test specimens, shall be utilized. No longer than 20 hours before testing, the fur of the animals shall be clipped on both sides of the spinal column. A vacuum shall be used for removal of the clipped hairs so as to prevent their introduction into the implantion site and thereby causing a tissue reaction. If necessary, the animals shall be anesthetized with a commonly used anesthetic agent to a degree deep enough to prevent muscular movements such as twitching. The specimens, of the specified size, shall be implanted by means of a sterile hypodermic needle (e.g., a 15 gage, 3/4 inch needle with I.V. point) and a sterile stylet. The needle may be presterilized and the sterile specimen aseptically inserted therein, or the needle with the clean specimen inserted in the cannula, and with the hub and cannula properly covered, shall be appropriately sterilized (allowing for proper degassing if agents such as ethylene oxide are used). Four specimens shall be implanted into the paravertebral site (e.g., spinus rectus or sacrospinalis muscle) of each of two rabbits. Each needle shall be inserted, at an angle of approximately 30 degrees, 2½ to 5 centimeters (cm.) from the midline and parallel to the spinal column, and approximately 2½ centimeters apart from each other. The sterile stylet shall be inserted into the needle to hold the specimen in the tissue while withdrawing the needle. If excessive bleeding is observed after implantation of a specimen, a duplicate shall be placed. After 14 days, the rabbits shall be sacrificed using common laboratory techniques (e.g., with an overdose of an anesthetic agent). To prevent the animals from bleeding, the autopsy shall be performed after a sufficient time has lapsed. The muscle tissue surrounding the implants shall then be examined, both grossly and microscopically, by a pathologist certified



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by the American Board of Pathology or the American College of Veterinary Pathologists. If no tissue reaction is noted, test is completed and material is acceptable (see 4.2.2.2). If tissue reaction is noted, a bacterial stain shall be utilized to help rule out contamination. If three or more of the tissue sites per animal show a reaction (not ruled out as contamination), test is completed and material is not acceptable (see 4.2.2.2). If one or two of the tissue sites per animal show a reaction (not ruled out as contamination), then retest on three more animals shall be performed (see 4.2.2.2 for acceptance criteria).

4.4.4 Test for lead. Ten (10) Gms. of plastic shall be taken, ignited to char, and then further ignited in a muffler at 700° C. to a white ash. The ash shall be boiled in 15 milliliters (ml.) of 10 percent nitric acid filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the current edition of the Pharmacopeia of the United States, shall be applied using a control of 5 p.p.m. or 12 p.p.m. of lead solution as given in the U.S.P. test. If the sample solution is darker than the limit, smaller aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in p.p.m. is determined.

4.4.5 Sterilization.

4.4.5.1 Steam sterilization. Tubes shall be steam autoclaved at 15 p.s.i.g. and 250° Fahrenheit (F) for 15 minutes and then allowed to cool at room temperature (68° F. to 72° F.).

4.4.5.2 Boil sterilization. Tubes shall be boiled in distilled water for 30 minutes and then allowed to cool at room temperature (68° F. to 72° F.).

4.4.5.3 Cold sterilization. Tubes shall be immersed in a chemical disinfectant for 20 minutes at room temperature (68° F. to 72° F.) and then followed by five (5) water rinses.

4.4.6 Hardness test. Hardness shall be determined in accordance with method 3021 of Federal Test Method Standard No. 601.



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## 5. PREPARATION FOR DELIVERY

### 5.1 Packaging.

5.1.1 Unit of issue. One (1) tube, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging quantities

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | 6 units | 360 units |

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit packaging. Each tube shall be packaged in a heat sealed, transparent, polyethylene envelope which shall be capable of maintaining sterility of contents as specified in 3.3. Each unit shall be further packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566, PPP-B-636, or PPP-B-676. Commercial colors shall be acceptable when carton conforming to PPP-B-566 or PPP-B-676 is utilized. Closure shall be adequate to prevent spillage of contents under normal handling.



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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; or PPP-B-621, table III, or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2. Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.3 Marking.

5.3.1 Interior packages. Each unit and intermediate package shall be marked in accordance with MIL-STD-129, except that when lables are utilized, waterproofing shall not be required. Lot (control) number, the legend "STERILE", contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) numbers shall be shown.



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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army. - MD
Navy. - MS
Air Force. - 03

Review activities:
Army. - MD
Navy. - MS
Air Force. - 03

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-T-36160A(DM) | TUBE, ENDOTRACHEAL, MAGILL (Plastic) |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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